

Student Performance Analysis Report

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Project Title: Student Performance Analysis Dashboard

1. OBJECTIVE

The objective of this project is to analyze student performance data and identify factors affecting academic results. The dashboard helps visualize student scores, gender distribution, parental education impact, and test preparation status.

2. DATASET DESCRIPTION

The dataset contains information about students including:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score

The dataset was imported into Power BI for analysis and visualization.

3. ANALYSIS PERFORMED

The following analyses were conducted:

- Student count analysis
- Average Math Score
- Average Reading Score

- Average Writing Score
- Gender-wise performance comparison
- Ethnicity-wise performance analysis
- Parental education impact on scores
- Test preparation course analysis

4. DASHBOARD FEATURES

The dashboard includes:

- KPI Cards
- Bar Charts
- Donut Chart
- Pie Chart
- Interactive Filters (Slicers)

These visualizations help users understand student performance trends effectively.

5. KEY FINDINGS

- Student performance varies across different demographic groups.
- Students who completed test preparation courses generally perform better.
- Parental education level has an impact on academic performance.
- Reading and writing scores are generally higher than math scores.

6. CONCLUSION

This project demonstrates how Power BI can be used to analyze educational data and generate meaningful insights. The dashboard provides an easy-to-understand visualization of student performance metrics and supports data-driven decision-making.

7. TOOLS USED

- Power BI Desktop
- CSV Dataset
- Microsoft Word

8. DASHBOARD SCREENSHOT

